

Project Title

Unsupervised Learning with Dimensionality Reduction and Clustering

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1. Abstract

This internship project explores the application of unsupervised machine learning techniques for the classification and visualization of handwritten digit images. Utilizing the Scikit-Learn `load_digits` dataset, which contains 1,797 samples of 8x 8 pixel grayscale images, the K-Means clustering algorithm is implemented to group similar structural patterns without prior label assistance. Principal Component Analysis (PCA) is used to compress the data into fewer dimensions while retaining the most important patterns, followed by a 10-cluster K-Means model to group the compressed features. Because unsupervised cluster indices are structurally arbitrary, a cluster-to-label translation pipeline was built using statistical majority-voting modes to map findings back to real digit classes (0–9). Finally, the system's performance is rigorously quantified using classification metrics including accuracy, a 10x10 confusion matrix, and precision-recall breakdowns.

2. Introduction

Handwritten digit recognition serves as a foundational benchmark in computer vision, pattern recognition, and machine learning. Automated character identification systems typically rely heavily on supervised neural networks. However, unsupervised configurations present an alternate paradigm: identifying core structural commonalities within raw pixel maps without historical annotation guidance. This project implements a clean ML pipeline to tackle unsupervised clustering on high-dimensional digital imagery, exploring how geometric and topological structures can be automatically organized. The algorithms used in the project work are:

- Principal Component Analysis (PCA): A linear orthogonal projection method that maps data along directions of maximal variance, reducing coordinate tracking space.
- K-Means Clustering: A distance minimization partitioning algorithm that isolates data patterns into distinct spherical segments around calculated cluster centroids.

During the first two weeks of internship, training was provided on the following topics:

- Python Basics (Data, Variables, Lists, Loops, Data Structures, Class, Functions, OOPS, NumPy, Pandas)
- Machine Learning (Foundations of ML & The Data Science Lifecycle, Supervised Learning for Continuous Variables: Linear Regression & Optimization, Supervised Learning for Categorical Outputs: Logistic Regression & Probabilistic Modeling, Unsupervised Data Exploration: Clustering and Dimensionality Reduction, Performance Evaluation, Error Analysis and the ML Lifecycle)
- LLM Fundamentals (LLM Guardrails, N-Gram Language Models, Tokenization, Embedding, Neural Networks/Transformers, Cosine Similarity, Word Embeddings, Supervised Fine Tuning)
- Communication Skills (Shannon Weaver's Model of Communication, Berlo's SMCR Model, Principles of Communication, Types of Communication)

3. Project Objective

The key objectives of this project are as follows:

- Pre-process and ingest 1,797 multi-dimensional handwritten digit samples from raw pixel frameworks.
- Isolate structural cluster centers across a baseline 64-dimensional workspace to verify input properties.
- Execute linear dimensionality reduction to compress the raw feature space down to 2 principal components for direct visual modeling mapping.
- Build an algorithmic cluster-to-target translation dictionary using majority statistical modes to label the unsupervised model output groups.
- Quantify the predictive limitations caused by heavy feature compression by generating an exact confusion matrix and a precision-recall classification report.

4. Methodology

Step 1: Data Ingestion and Validation

The pipeline starts by importing the `load_digits` from `sklearn.datasets`. This dataset holds 1,797 samples of handwritten characters formatted as 64-element flattened arrays.

Step 2: Baseline Centroid Extraction

Before compressing the feature space, a baseline K-Means model is fitted to the full 64-dimensional dataset to track the uncompressed cluster centers. The flat array shapes are refactored into descriptive spatial structures to analyze the cluster centers visually.

Step 3: Dimensionality Reduction via Principal Component Analysis (PCA)

To simplify computational tracking and enable downstream 2D space grouping, the 64 feature dimensions are mapped down to 2 distinct components using linear projection.

Step 4: Clustering Compressed Vectors

A separate K-Means model is initialized with 10 target groups and trained directly on the newly generated 2-dimensional spatial coordinate pairs.

Step 5: Label Mapping Pipeline

Each cluster ID was mapped to a real world digit class because unsupervised algorithms output arbitrary numerical cluster tags. This was done by calculating the statistical mode of the ground-truth targets within each isolated cluster group.

5. Data Analysis and Results

Array Dimension Profiles

The shapes of the preprocessing matrices at key stages of the pipeline are summarized below:

Operation Metric Checked	Resulting Matrix Shape	Strategic Implication
digits.data.shape	(1797,64)	The complete baseline dataset containing 1,797 samples and 64 features.
kmeans.cluster_centers_.shape	(10,64)	10 distinct cluster centers tracking the full 64 features.
centers[0].shape	(8,8)	Flat array shapes successfully refactored into a standard spatial grid.
reduced_data.shape	(1797,2)	The compressed dataset after mapping features to 2 coordinates.

Cluster Translation Dictionary Analysis

The translation dictionary reveals a major structural issue caused by the heavy 2D data compression:

{0: 2, 1: 0, 2: 4, 3: 8, 4: 1, 5: 8, 6: 9, 7: 6, 8: 7, 9: 0}

The target numbers 3 and 5 are completely missing from the dictionary values. The images representing 3 and 5 were heavily warped and overlapped with other numbers because significant structural variance is removed by the 2D PCA. As a result, they were completely absorbed by neighbouring clusters (primarily cluster 3 and cluster 5, which both mapped to a majority of digit 8).

Unsupervised System Accuracy

When the unsupervised cluster labels are converted into actual digit predictions using the dictionary map, the overall accuracy was evaluated as follows:

```
[9] ✓ 0s ▶ from sklearn.metrics import accuracy_score
      accuracy=accuracy_score(digits.target, pred_labels)
      print(accuracy)
      ... 0.5664997217584864
```

The pipeline delivers a system accuracy of 56.65%. While this performance is modest compared to supervised methods, it represents a reasonable baseline given that the data was compressed from 64 dimensions down to just 2 coordinates before clustering.

Confusion Matrix Evaluation

The confusion matrix provides a detailed look at where the unsupervised model gets confused:

```
[10]
✓ 0s from sklearn.metrics import confusion_matrix
model_confusionMatrix=confusion_matrix(digits.target, pred_labels)
print(model_confusionMatrix)
```

```
... [[167  0  0  0  1  0  7  0  3  0]
      [ 2 97  0  0  4  0  0  0 11 68]
      [ 0  3 114  0  0  0  0  0 10 47]
      [ 2  0 63  0  0  0  0  0 13 15]
      [ 0 11  0  0 138  0 27  3  2  0]
      [14  6  4  0  2  0  4 23 104 25]
      [22  0  0  0 21  0 136  0  2  0]
      [ 0 35  0  0  1  0  0 115 28  0]
      [ 0 10  3  0  1  0  0  24 136  0]
      [ 4  1  8  0  0  0  1  9  42 115]]
```

- Digit 0 (167 correct) and Digit 4 (138 correct) show excellent structural separation on the 2D plane.
- True Digit 3 was mostly misclassified as Digit 9 (90 samples) or Digit 2 (63 samples). True Digit 5 was heavily absorbed by the Digit 8 cluster (104 samples), leading to 0 correct direct mappings.

Classification Performance Report

The detailed metrics across all ten target classes are broken down below:

```
[11]
✓ 0s from sklearn.metrics import classification_report
print(classification_report(digits.target, pred_labels))
```

```
... precision    recall  f1-score   support
```

0	0.79	0.94	0.86	178
1	0.60	0.53	0.56	182
2	0.59	0.64	0.62	177
3	0.00	0.00	0.00	183
4	0.82	0.76	0.79	181
5	0.00	0.00	0.00	182
6	0.78	0.75	0.76	181
7	0.55	0.64	0.59	179
8	0.30	0.78	0.44	174
9	0.49	0.64	0.56	180
accuracy			0.57	1797
macro avg	0.49	0.57	0.52	1797
weighted avg	0.49	0.57	0.52	1797

- The system triggers an UndefinedMetricWarning for classes 3 and 5. Because no cluster maps to 3 or 5 as a majority mode, the precision and recall scores for these categories drop to 0.00, which pulls down the overall macro average.
- Conversely, digit 0 recorded the strongest f1-score of 0.86, followed closely by digit 4 at 0.79.
- Digit 8 shows a statistical anomaly with a recall of 0.78 and precision of 0.30. The model successfully found 78% of all real 8s in the dataset. The precision of 0.30

suggests that out of everything the model claimed was an “8”, only 30% actually were.

6. Conclusion

This project implementation shows that an unsupervised pipeline can successfully organize complex handwritten imagery into distinct structural groups without manually added labels. Forcing the data through a tight 2D PCA bottleneck makes it easy to visualize and computationally fast, but it drops the system accuracy to 56.65%. The fact that digits 3 and 5 were completely missed highlights that linear compression struggles to separate classes with highly overlapping pixel patterns.

Recommendations for Future Work

- Incorporate Non-Linear Reduction: Swap out standard PCA for non-linear embedding tools like t-SNE or UMAP to better preserve spatial margins between complex classes.
- Use Automated Cluster Selection: Use the Elbow Method or Silhouette Coefficient plots to find the optimal cluster count dynamically, rather than forcing a fixed $k=10$ structure.
- Feature Extraction Adjustments: Pass the raw image matrices through an autoencoder or a basic convolutional layer to extract structural features before clustering, rather than relying on flat pixel values.

7. APPENDICES

1. References - <https://drive.google.com/file/d/1yuQWgtaGmqDcn5RPNxb-TUqippZdr0oB/view?usp=drivesdk> ,
<https://docs.google.com/presentation/d/1DyNGANoAIYMdDMmKxI4oN2V3feoxQxoW/edit?usp=sharing&ouid=100095237014285084733&rtpof=true&sd=true> ,
https://docs.google.com/presentation/d/1-4giPXdG6F0Ck9Y7ewOqe0GaRV5ZDh3_/edit?usp=sharing&ouid=100095237014285084733&rtpof=true&sd=true .
2. Survey Questionnaire - Not Applicable.
3. https://github.com/Code-Queen06/IDEAS-SUMMER-INTERNSHIP_2026 -
Github Link.
4. Video Explanation -
<https://drive.google.com/file/d/1FJzeqx0pSjHAqNf21xad1BxzDQPuTYmi/view?usp=drivesdk>

